

Enterprise BI & Sales Performance Analytics

Power BI Portfolio Project Documentation

End-to-End Business Intelligence Reporting Project

Document item	Value
Project	Enterprise BI & Sales Performance Analytics
Platform	Microsoft Power BI Desktop
Dataset	Microsoft AdventureWorks Sales sample
Primary focus	Sales, profitability, customer, product and territory analysis
Output	Interactive multi-page Power BI report
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Status	Portfolio Project

1. Executive Summary

This project presents an end-to-end Business Intelligence solution developed in Microsoft Power BI to transform transactional sales data into an interactive management reporting environment. The objective is to provide decision-makers with a clear view of revenue, sales volume, profitability, product performance, customer activity and geographic performance.

The solution uses a dimensional data model, reusable DAX measures, interactive slicers and a set of business-oriented visualizations. The report is designed to move from high-level KPIs to detailed analysis, allowing users to identify performance trends and investigate the drivers behind revenue and profit.

1.1 Business Objective

The business objective is to create a reliable analytical report that answers questions such as:

- How much revenue is being generated?
- How much gross profit is being generated?
- How is profitability changing over time?
- Which products and categories contribute most to profit?
- Which countries or territories generate the strongest results?
- How do sales volume and profitability differ across products?
- Where should management investigate further?

1.2 Expected Business Value

The report consolidates sales and profitability information into a single analytical interface. Instead of relying on disconnected tables or manual calculations, users can filter the same set of measures by time, product and geography and immediately observe the effect on KPIs and charts.

2. Project Scope

The project covers the preparation, modeling, analysis and visualization of the AdventureWorks sales dataset in Power BI. The main analytical domains are:

Domain	Analysis
Sales	Revenue, orders, units sold, average order value and sales trends
Profitability	Costs, gross profit, profit margin and profit growth
Products	Category, subcategory and product-level performance
Geography	Country/territory-level sales and profitability
Time	Fiscal year/date analysis and year-over-year comparisons

2.1 Out of Scope

This portfolio project does not claim to represent a production enterprise deployment. Production security administration, automated cloud refresh infrastructure, enterprise gateway management and formal governance processes are outside the current scope unless separately implemented.

3. Dataset and Source

The project uses Microsoft's AdventureWorks Sales sample. The workbook contains related sales, product, customer, territory and date information suitable for demonstrating dimensional modeling and Power BI analytics.

Official Microsoft sample source:

<https://github.com/microsoft/powerbi-desktop-samples/raw/refs/heads/main/AdventureWorks%20Sales%20Sample/AdventureWorks%20Sales.xlsx>

Microsoft dimensional modeling guidance:

<https://learn.microsoft.com/en-us/power-bi/create-reports/desktop-dimensional-model-report>

3.1 Data Preparation

1. Open Power BI Desktop.
2. Select Get Data → Excel.
3. Load the AdventureWorks workbook.
4. Select the analytical tables rather than unnecessary helper/data sheets.
5. Open Transform Data to inspect data types, column names and data quality.
6. Confirm date fields are recognized as dates and numeric fields are numeric.
7. Remove unnecessary columns where appropriate.
8. Load the prepared tables into the Power BI model.

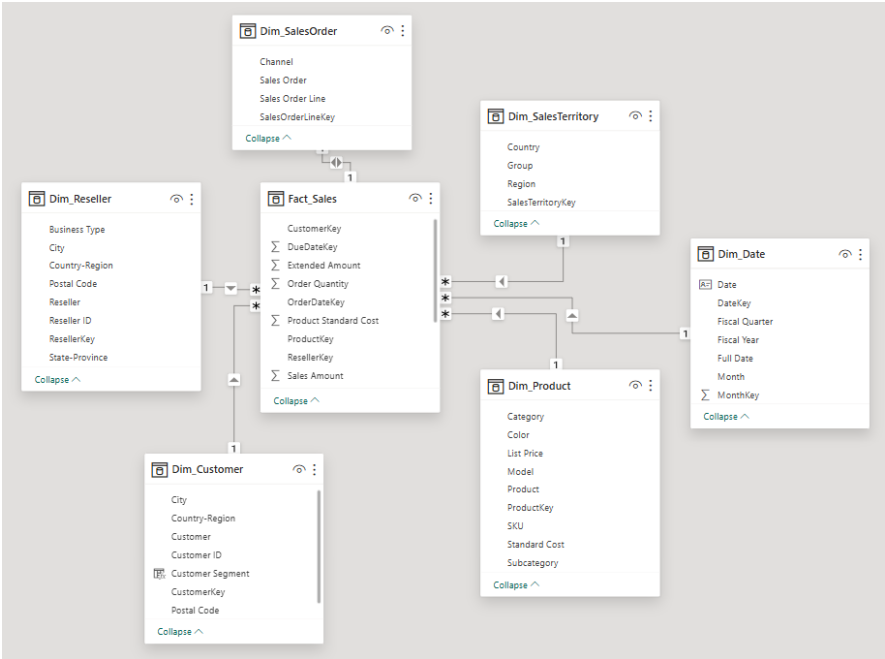
3.2 Data Quality Checks

- Check for blank or invalid keys.
- Check that sales amounts and costs are numeric.
- Check that quantities are numeric and non-negative where appropriate.
- Check date fields for valid dates.
- Check that dimension keys match the corresponding fact-table keys.
- Check for duplicate dimension keys.
- Validate the relationship between sales transactions and dimensions.

4. Data Model

The report is designed around a star-schema approach. The central fact table contains sales transactions, while descriptive dimensions provide the context required for analysis.

Conceptual model:



4.1 Fact Table

Fact_Sales is the central transactional table. Its grain should be documented as one sales-order line/product line per transaction. Typical analytical fields include Sales Order Line, SalesOrderLineKey, OrderDateKey, ProductKey, CustomerKey, SalesTerritoryKey, Order Quantity, Unit Price, Sales Amount and Product Standard Cost.

4.2 Dimension Tables

Dimension	Purpose	Example attributes
Dim_Date	Time analysis	Date, Year, Fiscal Year, Month
Dim_Product	Product analysis	Product, Product Number, Color, Size, List Price
Dim_SalesOrder	Orders analysis	Sales Order Line
Dim_Reseller	Reseller analysis	Business Type
Dim_Customer	Customer analysis	Customer attributes
Dim_SalesTerritory	Geographic analysis	Country, territory attributes

4.3 Relationships

The preferred relationship pattern is one-to-many from each dimension to Fact_Sales, with the dimension acting as the filtering side. Relationships should be active and use appropriate key columns.

4.4 Model Validation Checklist

- Dim_Date has a unique date/key column.
- Dimension-to-fact relationships are active.
- The fact table is on the many side.
- There are no accidental many-to-many relationships.
- Filters propagate from dimensions to Fact_Sales.

- The date dimension is used for time-intelligence calculations.

5. DAX Measures

Reusable DAX measures are used instead of repeating calculations inside individual visuals. This centralizes business logic and ensures consistent results throughout the report.

TotalSales

Total revenue generated from sales.

`TotalSales = SUM(FactSales[SalesAmount])`

TotalCosts

Total product costs.

`TotalCosts = SUM(FactSales[TotalProductCost])`

GrossProfit

Revenue minus product costs.

`GrossProfit = [TotalSales] - [TotalCosts]`

Profit Margin %

Gross profit as a percentage of revenue.

`Profit Margin % = DIVIDE([GrossProfit], [TotalSales])`

Total Orders

Number of distinct sales orders.

`Total Orders = DISTINCTCOUNT(FactSales[SalesOrderNumber])`

Unit Sold

Total quantity sold.

`UnitSold = SUM(FactSales[OrderQuantity])`

Average Order Value

Average revenue per order.

`Average Order Value = DIVIDE([TotalSales], [Total Orders])`

Sales LY

Revenue for the equivalent prior-year period.

`Sales LY = CALCULATE([TotalSales], SAMEPERIODLASTYEAR(DimDate[Date]))`

YoY Growth %

Year-over-year revenue growth.

`YoY Growth % = DIVIDE([TotalSales] - [Sales LY], [Sales LY])`

Profit LY

Gross profit for the equivalent prior-year period.

Profit LY = CALCULATE([GrossProfit], SAMEPERIODLASTYEAR(DimDate[Date]))

Profit YoY Growth %

Year-over-year gross-profit growth.

Profit YoY Growth % = DIVIDE([GrossProfit] - [Profit LY], [Profit LY])

Important time-intelligence note: Profit YoY Growth % requires an appropriate date context. When no specific fiscal year/date period is selected, there may be no single current period to compare against one prior-year period, so a card can display BLANK. Selecting a specific fiscal year provides the necessary comparison context.

Average order per customer

Average number of orders per customer

Average order per customer = DIVIDE ([Total Orders], [TotalCustomers])

Customer Growth %

Year-over-year customers growth.

Customer Growth % = DIVIDE([TotalCustomers] - [CustomerLY], [CustomerLY])

Sales per Customer

Average sales for each customer

Sales per Customer = DIVIDE([TotalSales], [TotalCustomers])

Sales YTD

Year-to-Date Sales

Sales YTD = TOTALYTD([TotalSales], Dim_Date[Date])

TotalCustomers

Total number of Customers

TotalCustomers = DISTINCTCOUNT(Fact_Sales[CustomerKey])

5.1 Formatting Standards

Measure	Recommended format
TotalSales	Currency / Millions / 2 decimal
GrossProfit	Currency / Millions / 2 decimal
TotalCosts	Currency / Millions / 2 decimal
Profit Margin %	Percentage / 2 decimal
Profit YoY Growth %	Percentage / 2 decimal

YoY Growth %	Percentage / 2 decimal
Unit Sold	Whole number
Total Orders	Whole number
AverageOrderValue	Currency / 2 decimal
Profit LY	Currency / Millions / 2 decimal
Sales LY	Currency / Millions / 2 decimal
Average order per customer	Whole number
Customer Growth %	Percentage / 2 decimal
Sales per customer	Currency / Millions / 2 decimal
Sales YTD	Currency / Millions / 2 decimal
Total Customers	Whole number

6. Design and Navigation Report

The report should follow a management-oriented flow: start with an overview, then move into sales performance, customer/product analysis and finally profitability analysis. Slicers should be positioned consistently across pages and should provide a clear filtering experience.

6.1 Recommended Report Pages

Page	Purpose
1. Executive Overview	High-level KPIs and overall sales performance
2. Product Analysis	Sales trends by product category, Costs, units sold and geographic/product performance
3. Customer Analysis	Customer distribution, sales and loyalty
4. Profitability Analysis	Gross profit, margin, profit trends and profitability drivers

7. Page 1 — Executive Overview

Purpose: provide management with an immediate summary of overall commercial performance.

Visual	Title	Fields / Measures	Configuration
Card	Total Sales	[TotalSales]	Currency, Millions, 1 decimal
Card	Gross Profit	[GrossProfit]	Currency, Millions, 1 decimal
Card	Profit Margin %	[Profit Margin %]	Percentage, 1 decimal
Card	Total Orders	[Total Orders]	Whole number
Line chart	Sales Trend	DimDate[Year-Month] + [TotalSales]	Sort Year-Month chronologically
Bar chart	Sales by Product Category	Category + [TotalSales]	Descending by sales
Bar chart	Sales by Country	Country + [TotalSales]	Descending by sales
Line chart	YoY Growth by Fiscal year	Fiscal Year + [YoY Growth]	Ascending Fiscal Year

8. Page 2 — Product Analysis

Purpose: Identify Products, Categories, and Sub-categories and their distribution as well as costs and revenues associated to each one.

Visual	Title	Fields / Measures	Configuration
Card	Total Sales	[TotalSales]	Currency
Card	Units Sold	[UnitSold]	Whole number
Card	Number of Sub-catrgories	Distinct count of subcatregories	Whole number
Card	Number of Products	Distinct count of Product	Whole number
Line chart	Units sold by fiscal year	Fiscal year + [Unit Sold]	Time series
Bar chart	Costs by ProductCategory	Category + [TotalCosts]	Descending
Bar chart	Sales by Product Category	Category + [TotalSales]	Descending
Bar chart	Top 5 Gross Profit by Product Sub-category	Product + [Gross Profit]	Top N = 5
Pie chart	Percentage of Profit Margin by Category	Category + [Profit Margin %]	Percentage of total
Matrix	Product Analytics	Product + [TotalSales] + [Unit Sold] + [Gross Profit] + [Profit Margin %]	-

9. Page 3 — Customer Analysis

Purpose: identify customer segments and distribution associated with sales activity.

Visual	Title	Fields / Measures	Configuration
Card	Total Customers	[TotalCustomers]	Whole number
Card	Total number of Orders	[Count of Sales Orders]	Whole number
Card	Average order per Customer	[Average order per Customer]	Whole number
Card	Average revenue per Customer	[Sales per Customer]	Currency
Bar chart	Top 5 Sales by customers	Customer + [Sales per Customer]	Top N = 5
Line chart	Percentage of Customer Growth by fiscal year	Fiscal Year + [Customer Growth %]	Time Series
Donut chart	Customer Segment	Customer Segment + [TotalCustomers]	Percentage of total
Map	Customers Distribution by continents	Country + [TotalCustomers]	Use the clearest geographic visual available
Matrix	Customer Analytics	Customer + [Total Orders] + [Average order per customer] + [Sales per Customer] + [Customer Segment]	-

10. Page 4 — Profitability Analysis

Purpose: explain where profit is generated, how profitability changes over time and which products/categories require further investigation.

Visual	Title	Fields / Measures	Configuration
Card	Total Revenue	[TotalSales]	Currency, Millions, 1 decimal
Card	Gross Profit	[GrossProfit]	Currency, Millions, 1 decimal
Card	Profit Margin %	[Profit Margin %]	Percentage, 1 decimal
Card	Profit YoY Growth %	[Profit YoY Growth %]	Percentage, 1 decimal; use with a selected fiscal year
Line chart	Revenue & Gross Profit Trend	Month + [TotalSales] + [GrossProfit]	Two measures over time
Clustered Bar chart	Gross Profit by Product Category	Category + [GrossProfit]	Descending
Clustered Bar chart	Gross Profit by Country	Country + [GrossProfit]	Descending
Bar chart	Top 10 Products by Gross Profit	Product + [GrossProfit]	Top N = 10
Scatter chart	Sales vs Profitability by Product	X=[TotalSales], Y=[Profit Margin %], Size=[GrossProfit], Details=Product	Use category as legend if available

10.1 Page 4 Slicers

Slicer	Field	Recommended behavior
Fiscal Year	DimDate[Fiscal Year]	Single select if using Profit YoY Growth card as a headline KPI
Product	DimProduct / Product field	Select products where required
Country	DimTerritory[Country]	Filter geographic performance
Category	DimProductCategory[CategoryName]	Filter product category

10.2 Profitability Matrix Conditional Formatting

- Apply data bars to Gross Profit.
- Use conditional formatting on Profit Margin %.
- Keep the number formats consistent with the KPI cards.
- Allow drill-down from Category → Subcategory → Product.

11. Step-by-Step Power BI Implementation

Step 1 — Import data

Open Power BI Desktop → Get Data → Excel → select the AdventureWorks Sales workbook → Transform Data → review tables → Close & Apply.

Step 2 — Build the model

Open Model view. Connect Dim_Date, Dim_Product, Dim_Customer and Dim_Territory to Fact_Sales using the appropriate keys. Ensure relationships are active and one-to-many.

Step 3 — Create a Measures table

Create a dedicated table named Measures_ for reusable DAX measures. Keep analytical measures in this table for easier report maintenance.

Step 4 — Create base measures

Create TotalSales, TotalCosts, Gross Profit, Profit Margin %, Total Orders, Unit Sold and Average Order Value.

Step 5 — Create time-intelligence measures

Create Sales LY and YoY Growth %. Then create Profit LY and Profit YoY Growth %. Test them with a specific fiscal year.

Step 6 — Format measures

Apply currency, percentage and whole-number formats. Use millions for large currency KPIs where appropriate.

Step 7 — Create report pages

Create the four report pages and maintain consistent page titles, slicers and visual spacing.

Step 8 — Add KPI cards

Place the key measures in cards at the top of each page. Keep the most important KPIs toward the upper-left area.

Step 9 — Add analytical visuals

Add time-series, bar/column, scatter and matrix visuals according to the page specifications.

Step 10 — Configure filters

Add Date, Fiscal Year, Product, Country, Category and other relevant slicers. Test cross-filtering between visuals.

Step 11 — Add interactions

Use Format → Edit interactions to ensure clicking a chart filters or highlights the intended visuals.

Step 12 — Validate numbers

Cross-check selected totals against the underlying data and inspect results under different year/category/country filters.

Step 13 — Improve usability

Align visuals, use consistent titles, reduce unnecessary borders, avoid excessive colors and ensure labels are readable.

Step 14 — Final QA

Test every slicer, drill-down, tooltip, measure and visual. Check that no visual shows errors or unexpected blanks.

12. Validation and Testing

Test	Expected result
Select one fiscal year	Profit YoY Growth % returns a numeric percentage when prior-year data exists.
Change country	Sales, profit and visuals respond to the selected country.
Change category	Product and profitability visuals respond to the selected category.
Select a product	Relevant KPIs and charts update consistently.
Clear all filters	Overall totals return to the report's full available dataset.
Check Gross Profit	Gross Profit equals Total Sales minus Total Costs.
Check Margin	Profit Margin equals Gross Profit divided by Total Sales.
Check AOV	Average Order Value equals Total Sales divided by Total Orders.

12.1 Specific YoY Test

When Fiscal Year is set to a single year, verify that Profit YoY Growth % compares the selected year's Gross Profit with the equivalent prior-year period. When Fiscal Year is set to All, a blank result can be appropriate because there is no single selected current period for a one-year comparison.

13. Design Guidelines

- Use descriptive visual titles that explain the business question.
- Keep KPI cards at the top of the page.
- Use line charts for time trends.
- Use bar/column charts for category comparisons.
- Use scatter plots for relationships between two quantitative measures.
- Use matrices for detailed investigation and drill-down.
- Avoid unnecessary visual variety.
- Maintain consistent number formatting.
- Keep slicers in a consistent position across pages.
- Use whitespace to separate logical sections.

14. Documentation of Business Logic

The analytical logic of the report is intentionally expressed through reusable measures. This makes the report easier to maintain and allows multiple visuals to use the same business definitions.

Business concept	Definition
Revenue	Sum of sales amount.
Cost	Sum of total product cost.
Gross Profit	Revenue minus product cost.
Profit Margin	Gross profit divided by revenue.
Total Orders	Distinct count of sales orders.
Units Sold	Sum of order quantity.

Average Order Value	Revenue divided by distinct orders.
Sales YoY Growth	Current-period revenue compared with equivalent prior-year period.
Profit YoY Growth	Current-period gross profit compared with equivalent prior-year period.

15. Final Deliverables Checklist

- ☐ Power BI PBIX report completed.
- ☐ Star-schema data model validated.
- ☐ Measures_ table created.
- ☐ Core DAX measures created and formatted.
- ☐ Four report pages completed.
- ☐ Slicers configured.
- ☐ Profit YoY Growth tested with a selected fiscal year.
- ☐ All visuals have meaningful titles.
- ☐ No broken visuals or unexpected errors remain.
- ☐ Screenshots captured for portfolio presentation.
- ☐ Project documentation exported as DOCX/PDF if required.

17. References

- Microsoft Power BI dimensional modeling guidance: <https://learn.microsoft.com/en-us/power-bi/create-reports/desktop-dimensional-model-report>
- Microsoft Power BI visualization overview: <https://learn.microsoft.com/en-us/power-bi/visuals/power-bi-visualizations-overview>
- Microsoft Power BI dashboard design tips: <https://learn.microsoft.com/en-us/power-bi/create-reports/service-dashboards-design-tips>
- Microsoft AdventureWorks Sales sample: <https://github.com/microsoft/powerbi-desktop-samples/raw/refs/heads/main/AdventureWorks%20Sales%20Sample/AdventureWorks%20Sales.xlsx>